

Lower-Sideband Scalar Balance and Target-Scale Underdetermination

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Abstract

At the mandatory lower sideband $2m = 2k - 2$, RH-346 gives the exact physical coefficient

$$p_{\sigma,k,2m} = Y_m^- + \mathcal{P}_{\sigma,2m} - S_m^-, \quad Y_m^- = \mathcal{T}_{k,m}^{\text{rest}} - d_{\sigma,k,2m}, \quad S_m^- = F_m^{\text{orb}} + \mathcal{A}_{k,2m}.$$

Here $m = k - 1$ is an orbit-period parameter on the same (σ, k) noise clock, $F_m^{\text{orb}} = 2mG_m$, and

$$\frac{S_m^-}{F_m^{\text{orb}}} \longrightarrow 1, \quad \frac{G_m}{H_m} = \frac{(\beta R)^{2m}}{C_M m} \{1 + o(1)\} \longrightarrow \infty.$$

The inherited parity interface is $\mathcal{P}_{\sigma,2m}/S_m^- \rightarrow C_* C_M \lambda^{\eta-1}$. Thus, if the actual signed remainder satisfies $Y_m^- = o(H_m)$, every fixed phase other than

$$\eta_- = 1 - \frac{\log(C_* C_M)}{\log \lambda}$$

has a divergent weighted lower contribution, with exact leading coefficient $|C_* C_M \lambda^{\eta-1} - 1|/C_M$.

At the balance phase we construct two exact scalar parity envelopes using the order- $2m$ inverse parity map. The choices $\mathcal{P}_m^{\text{close}} = S_m^-$ and $\mathcal{P}_m^{\text{far}} = S_m^- + F_m^{\text{orb}}/m$ both lie in the legal domain eventually and satisfy the same square-root law on the same clock with $k = m + 1$. In the scalar information class with $Y_m^- = 0$, they give zero residual versus $F_m^{\text{orb}}/m = 2G_m$, whose weighted value is exactly $G_m/H_m \rightarrow \infty$. These scalar sequences are not noisy operators. Actual lower compensation, the remaining off-alias aggregate, determinant gluing, and Gates A–E remain open. No Riemann-hypothesis conclusion follows.

1 Exact lower-sideband scalar ledger

Use the physical clock and target

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_\sigma = k - \frac{\log(1/\sigma)}{2 \log \lambda}, \quad m = k - 1, \quad H_m = mR^{-2m}, \quad (1)$$

where $R = 7/5$. The replacement $m = k - 1$ does not define a second noise sequence. It only identifies the period parameter of the lower boundary orbit.

RH-346 completes that orbit extraction and proves the exact direct identity

$$\boxed{p_{\sigma,k,2m} = Y_m^- + \mathcal{P}_{\sigma,2m} - S_m^-, \quad Y_m^- = \mathcal{T}_{k,m}^{\text{rest}} - d_{\sigma,k,2m}, \quad S_m^- = F_m^{\text{orb}} + \mathcal{A}_{k,2m}.} \quad (2)$$

The two deterministic constituents are

$$F_m^{\text{orb}} = 2mG_m, \quad G_m = \frac{r_H^{-2m}}{1 + |M_m|}, \quad (3)$$

$$\mathcal{A}_{k,2m} = 2(\beta^{2m} - \beta_k^{2m}), \quad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (4)$$

The radial sideband need not have a source-locked sign. What is proved is the relative law

$$\frac{\mathcal{A}_{k,2m}}{F_m^{\text{orb}}} = \frac{C_M - 1}{m} + o(m^{-1}) \longrightarrow 0. \quad (5)$$

Consequently $S_m^- > 0$ eventually and

$$S_m^- = \frac{2m}{C_M} \beta^{2m} \{1 + o(1)\}, \quad (6)$$

$$\frac{S_m^-}{H_m} = \frac{2}{C_M} (\beta R)^{2m} \{1 + o(1)\} \longrightarrow \infty, \quad (7)$$

$$\frac{G_m}{H_m} = \frac{1}{C_M m} (\beta R)^{2m} \{1 + o(1)\} \longrightarrow \infty. \quad (8)$$

The strict inequality $\beta R > 1$ is already certified in the source chain [2, 6].

The actual even parity packet is

$$\mathcal{P}_{\sigma,2m} = r_H^{-2m} \{1 - (1 - \delta_\sigma)^{2m}\}, \quad \delta_\sigma = C_* \sqrt{\sigma} + o(\sqrt{\sigma}), \quad C_* > 0. \quad (9)$$

On bounded physical phase, RH-346 proves

$$\frac{\mathcal{P}_{\sigma,2m}}{S_m^-} = C_* C_M \lambda^{\eta_\sigma - 1} \{1 + o(1)\}. \quad (10)$$

The factor λ^{-1} is forced by keeping $k = m + 1$ in (1); it is not a change of normalization [4, 5].

2 Conditional obstruction away from scalar balance

Put

$$\gamma_-(\eta) = C_* C_M \lambda^{\eta - 1}. \quad (11)$$

Since $C_*, C_M > 0$ and $\lambda > 1$, this function is strictly increasing. Its unique leading balance phase is

$$\boxed{\eta_- = 1 - \frac{\log(C_* C_M)}{\log \lambda}}, \quad \gamma_-(\eta_-) = 1. \quad (12)$$

This symbolic interface is inherited from RH-346; it is not the new theorem of the present paper.

Theorem 2.1 (Conditional physical off-balance obstruction). *Consider an actual physical sequence on (1) with $\eta_\sigma \rightarrow \eta$. If*

$$Y_m^- = o(H_m) \quad (13)$$

and $\eta \neq \eta_-$, then

$$\boxed{\frac{|p_{\sigma,k,2m}|}{2H_m} = \frac{|C_* C_M \lambda^{\eta - 1} - 1|}{C_M} (\beta R)^{2m} \{1 + o(1)\} \longrightarrow \infty.} \quad (14)$$

Equivalently, the single weighted prefix contribution $|p_{\sigma,k,2m}| R^{2m} / (2m)$ diverges.

Proof. By (7), hypothesis (13) gives $Y_m^- / S_m^- \rightarrow 0$. Divide (2) by S_m^- and use (10):

$$\frac{p_{\sigma,k,2m}}{S_m^-} \longrightarrow \gamma_-(\eta) - 1 \neq 0.$$

Moreover (6) gives

$$\frac{S_m^-}{2H_m} = \frac{1}{C_M} (\beta R)^{2m} \{1 + o(1)\}.$$

Multiplication proves (14). The weighted identity follows from $H_m = m R^{-2m}$. $\square$

Remark 2.2 (Scope of the hypothesis). The remainder Y_m^- contains the actual signed orbit-free raw contribution and the actual head defect. No repository theorem proves (13). Therefore theorem 2.1 excludes parity as an isolated scalar compensation mechanism under a named physical hypothesis; it does not prove actual lower-sideband or aggregate prefix nonclosure.

3 Balance requires exponentially sharper information

At $\eta = \eta_-$, the source law (10) has leading ratio one, but target closure requires more than a relative $o(1)$ estimate.

Proposition 3.1 (Necessary balance-phase precision). *Under (13), lower-sideband closure is equivalent to*

$$\mathcal{P}_{\sigma,2m} = S_m^- + o(H_m). \quad (15)$$

Relative to the positive demand, the required precision is

$$o\left(\frac{H_m}{S_m^-}\right) = o((\beta R)^{-2m}). \quad (16)$$

The inherited relative $o(1)$ in (10) does not decide (15).

Proof. Rearrange (2) and use (13). Formula (7) gives (16). An unspecified vanishing relative error need not be exponentially small. $\square$

Ordinary substitution of archived decimal constants gives $\eta_- = 4.0609149137\dots$. Those decimals are reproduction inputs, not directed interval enclosures. In particular, this numerical value cannot be used as a rigorous theorem excluding the balance phase from the usual canonical clock window. The analytic argument below therefore treats the symbolic balance phase directly.

4 Exact scalar underdetermination at balance

Freeze the same physical clock at the symbolic balance:

$$k = m + 1, \quad \sigma_m = \lambda^{-2(k-\eta_-)}. \quad (17)$$

Using (12),

$$C_*\sqrt{\sigma_m} = \frac{1}{C_M}\lambda^{-m}. \quad (18)$$

For a desired scalar packet $0 < X_m < r_H^{-2m}$, define the exact inverse parity map

$$\delta_m(X) = 1 - \{1 - r_H^{2m}X_m\}^{1/(2m)}, \quad \lambda_{-,m}(X) = -(1 - \delta_m(X)). \quad (19)$$

Then $\lambda_{-,m}(X) \in (-1, 0)$ and substitution in (9) recovers X_m exactly.

Theorem 4.1 (Opposite exact lower scalar completions). *For all sufficiently large m , let*

$$X_m^{\text{close}} = S_m^-, \quad X_m^{\text{far}} = S_m^- + \frac{F_m^{\text{orb}}}{m}. \quad (20)$$

Both packets satisfy the legal domain

$$0 < r_H^{2m}X_m^{\text{close}} < 1, \quad 0 < r_H^{2m}X_m^{\text{far}} < 1, \quad (21)$$

and their inverse images obey the same leading physical law

$$\delta_m(X^{\text{close}}) = C_*\sqrt{\sigma_m}\{1 + o(1)\}, \quad \delta_m(X^{\text{far}}) = C_*\sqrt{\sigma_m}\{1 + o(1)\}. \quad (22)$$

In the scalar information class with $Y_m^- = 0$, the close sequence has $p_{\sigma,k,2m} = 0$, whereas the far sequence has

$$p_{\sigma,k,2m} = \frac{F_m^{\text{orb}}}{m} = 2G_m \quad (23)$$

and the exact weighted identity

$$\frac{|p_{\sigma,k,2m}|}{2H_m} = \frac{G_m}{H_m} = \frac{1}{C_M m}(\beta R)^{2m}\{1 + o(1)\} \longrightarrow \infty. \quad (24)$$

Proof. Equations (6) and $\beta r_H = \lambda^{-1/2}$ give

$$r_H^{2m} S_m^- = \frac{2m}{C_M} \lambda^{-m} \{1 + o(1)\} \longrightarrow 0. \quad (25)$$

Likewise

$$r_H^{2m} \frac{F_m^{\text{orb}}}{m} = \frac{2}{C_M} \lambda^{-m} \{1 + o(1)\} \longrightarrow 0.$$

Eventual positivity of S_m^- proves (21). For $x_m \rightarrow 0$,

$$1 - (1 - x_m)^{1/(2m)} = \frac{x_m}{2m} \{1 + o(1)\}. \quad (26)$$

Apply this with $x_m = r_H^{2m} X_m$. The far correction is smaller than S_m^- by a factor $m^{-1} \{1 + o(1)\}$, so both choices give

$$\delta_m(X) = \frac{1}{C_M} \lambda^{-m} \{1 + o(1)\} = C_* \sqrt{\sigma_m} \{1 + o(1)\},$$

proving (22). Exact recovery by (19) and substitution in (2) with $Y_m^- = 0$ give zero and $F_m^{\text{orb}}/m = 2G_m$. Dividing the latter by $2H_m$ proves the exact first equality in (24); (8) completes the proof. $\square$

Remark 4.2 (Information class, not physical realization). The map (19) constructs scalar eigenvalue sequences satisfying the exact packet formula and the leading square-root law. It does not construct noisy transfer operators, replace the actual $\lambda_-(\sigma)$, or prescribe the actual orbit-free remainder. The opposite target behaviors prove insufficiency of the scalar information class only.

5 Executable protocol, verdict, and next route

The executable artifact evaluates the exact period- $2m$ boundary multiplier, the period- $2k$ radial sideband, and the two inverse parity envelopes at $m = 8, 16, 24, 40$ with $k = m + 1$. It checks:

1. the same-clock relation and the exact phase equation;
2. positivity and the legal inverse-map domain for both packets;
3. exact recovery at exponent $1/(2m)$;
4. convergence of both gap ratios to the common square-root law;
5. zero versus $F_m^{\text{orb}}/m = 2G_m$ scalar residuals; and
6. equality of the far weighted term with G_m/H_m .

The finite rows are formula-reproduction checks only. They are not interval certificates, asymptotic evidence, or observations of an actual noisy operator.

The lower scalar route is therefore **STOP_SCOPED**: off balance it fails under a target-negligible actual remainder, and at balance the proved scalar information admits opposite exact target behavior. Actual lower compensation and noncompensation remain **NOT_TESTABLE/open**.

RH-348 must return to the punctured strict-prefix aggregate

$$2 \leq n < 4k, \quad n \notin \{2k, 2k - 2\},$$

and either control it as one signed physical object or isolate another physical sideband atom. Two selected-order analyses do not close $E_{\text{off},(4k)}$ [1, 3, 7].

RH-288 remains inactive. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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